

Heaven's Light is Our Guide



Department of Electrical & Computer Engineering
Rajshahi University of Engineering & Technology

Course Plan / Outline

1. **Course Information:**

Course Title	Co-ordinate Geometry and Partial Differential Equations
Course Code	Math 2217
Course Type	Basic Science and Mathematics (Theory)
Credit	3.00 (Three Contact Hours per Week), three lecture slots per week according to class routine.
Semester	2 nd Year Even Semester

2. **Instructor Information:**

Name of the Instructor	To be announced before starting of the semester
Designation	
Office Room	
Cell	
Email	
Webpage link	

3. **Prerequisite (if have any): None**

4. **Course Rationale:**

This course aims to equip students with the necessary knowledge and skills in co-ordinate geometry and partial differential equations, which are fundamental concepts in many areas of mathematics, science, and engineering. The course covers topics such as lines, circles, conic sections, parametric equations, surfaces, and quadric surfaces. Additionally, the course covers partial differential equations, including methods of solution, boundary value problems, and applications in physics and engineering. Through this course, students will gain a solid foundation in the mathematical tools needed to tackle complex problems and analyze real-world phenomena in various fields, such as physics, engineering, and economics.

5. **Course Objectives:**

After successfully completing this course, students will-

1. Understand and apply the fundamental concepts of three-dimensional coordinate geometry, including system of coordinates, transformations, distance, and equations of lines and planes.
2. Develop proficiency in solving first-order partial differential equations using Lagrange's method and finding integral surfaces passing through a given curve.
3. Be proficient in solving non-linear partial differential equations of order one, including complete, particular, singular, and general integrals using Charpit's method.
4. Develop proficiency in solving second-order partial differential equations through separation of variables and identifying the nomenclature and classifications of standard parabolic, elliptic, and hyperbolic forms.
5. Gain an understanding of series solutions to differential equations and the properties of Bessel's functions and Legendre's polynomials.

6. **Course Content / Course Description:**

Co-ordinate Geometry: Coordinate geometry of three dimension- System of coordinates, transformations of coordinates, distance between two points, section formula, projection, direction cosines, equations of planes and lines.

Partial differential equations: Four rules for solving simultaneous equations of the form $P=Q=r$; Lagrange's method of solving PDE of order one, Integral surfaces passing through a given curve, Non-linear PDE of order one (complete, Particular, Singular and general integrals), Charpit's method, Second order PDE; Its nomenclature and classifications to canonical (Standard) parabolic, elliptic, hyperbolic, Solution by separations of variables, Linear PDE with constants coefficients.

Series solution: Solution of differential equations in series by the method of Frobenius, Bessel's functions, Legendre's Polynomials and their properties.

7. **Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)**

CLO No.	Course Learning Outcomes (CLO)	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	Analyze the principles of three-dimensional coordinate geometry to accurately describe points, lines, planes, and transformations in space, as well as calculate distances, projections, and direction cosines.		C4										
CLO2	Solve various types of partial differential equations, including linear PDEs with constant coefficients and second-order PDEs, using methods such as Lagrange's method, Charpit's method, and separation of variables.	C3											
CLO3	Apply the method of Frobenius, Bessel's functions, and Legendre's Polynomials to solve differential equations with series solutions, and understand the properties of these solutions, including their convergence and divergence.	C3											

8. **CLOs Assessment Policy:**

CLOs	Class Test	Assignment	Project	Presentation	Final Exam	Others
CLO1	√				√	
CLO2	√				√	

CLO3	√	√			√	
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9. **References:**

SL No.	Text Book/Reference Book(s)	Author(s)	Online Availability
1.	Analytical Co-Ordinate Geometry	S. I. Loney	Link is provided in Google Class Room
2.	A text book on Coordinate Geometry with Vector Analysis	Rahman & Bhattacharjee	Link is provided in Google Class Room
3.	Differential Equations	L. S. Ross	Link is provided in Google Class Room
4.	Partial Differential Equations	Schamum's Outline Series	Link is provided in Google Class Room
5.	Advanced Engineering Mathematics	H.K. Dass	Link is provided in Google Class Room

10. **Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning assessment strategy mapped with CLOs**

Week	Topic	Delivery Material	Teaching Learning Strategy	Assessment Strategy	Aligned CLOs
1	Co-ordinate Geometry: Coordinate geometry of three dimension- System of coordinates	Books	☒ Lecture ☐ Tutorial ☐ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO1
2	Transformations of coordinates, distance between two points	Books	☒ Lecture ☐ Tutorial ☐ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO1
3	Section formula, projection, direction cosines, equations of planes and lines.	Books	☒ Lecture ☐ Tutorial ☐ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO1
4	Partial differential equations: Four rules for solving simultaneous equations of the form $P=Q=r$	Books	☒ Lecture ☐ Tutorial ☐ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO2
5	Lagrange's method of solving PDE of order one,	PPT slide	☒ Lecture ☐ Tutorial ☐ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO2
6	Integral surfaces passing through a given curve	PPT slide	☒ Lecture ☐ Tutorial ☐ Discussion ☐ Interaction	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation	CLO2

			☐ Audio/Video ☐ Others	☐ Project ☐ Others	
7	Non-linear PDE of order one (complete, Particular, Singular and general integrals)	Lecture Notes, Books	☒ Lecture ☐ Tutorial ☐ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO2
8	Charpit's method, Second order PDE; Its nomenclature and classifications to canonical (Standard) parabolic, elliptic, hyperbolic	Books	☒ Lecture ☐ Tutorial ☐ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO2
9	Solution by separations of variables, Linear PDE with constants coefficients.	PPT Slide, Lecture Notes	☒ Lecture ☐ Tutorial ☐ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO2
10-12	Series solution: Solution of differential equations in series by the method of Frobenius, Bessel's functions, Legendre's Polynomials and their properties.	PPT Slide	☒ Lecture ☐ Tutorial ☐ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☒ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	CLO3
13	Review	PPT Slide	☒ Lecture ☐ Tutorial ☐ Discussion ☐ Interaction ☐ Audio/Video ☐ Others	☒ Class Test ☐ Assignment ☒ Final Exam ☐ Presentation ☐ Project ☐ Others	

11. Virtual Availability of Lecture Materials:

All resources related to this course are available in online platform.

Medium	Google classroom/Zoom/MS Teams etc.
Instructions	
Class Code	

12. Assessment & Marks Distribution:

SL No.	Name of Assessment		Marks	Date
1.	Continuous Assessment	Class Participation and Attendance	10	As Per Class Routine
2.		Class Test	20	As Per Class Test Routine
3.		Assignment/Project/Viva-voce/Presentation etc.	10	Mentioned by the Instructor
4.	Final Examination		60	As Per Examination Routine
Total Marks:			100	

13. **Grading Policy:**

Mark wise grading policy of this course is as follows:

% of Marks	Letter Grade	Point
80 – 100	A+	4.00
75 – 79	A	3.75
70 – 74	A-	3.50
65 – 69	B+	3.25
60 – 64	B	3.00
55 – 59	B-	2.75
50 – 54	C+	2.50
45 – 49	C	2.25
40 – 44	D	2.00
0 – 39	F	0.00

14. **Additional Course Policy:**

SL. No.	Name of the Test	Description
1.	Class Participation and Attendance	Attending classes are mandatory for the students. Students will be marked in a scale of 10 marks according to Academic Ordinance for the Undergraduate Studies.
2.	Class Test	For this course, four class tests will be held and marks of each class test is 20. Best three class tests average mark is counted as the score of class test. Any unfair means is strictly prohibited in the examination hall.
3.	Assignment/Project/Viva-voce/Presentation etc.	All the assessment strategies are considered as regular assessment. The course teacher will assign works to the students and evaluate them with a scale 10 marks that will be counted as regular assessment marks in final scoring.
4.	Semester Final	For this course, semester final examination will be held according to examination routine at end of this semester. Semester final examination contains 60 marks. Any unfair means is strictly prohibited in the examination hall.

15. **Students Policy with Physically Handicapped:**

Students with disabilities are required to inform the Department of Computer Science & Engineering of any specific requirement for classes or examination as soon as possible

16. **Generic Skills (Indirect):**

SL. No.	Generic skills	Attained Generic Skills (Please tick against the relevant skill)	How the skills will be developed
1.	Presentation skill		
2.	Writing skill		
3.	Interpersonal skill		
4.	Leadership skill		
5.	Community skills		

17. **Quick Links:**

- Website: <https://www.ece.ruet.ac.bd>